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Super-Chandrasekhar white dwarfs supported by a purely toroidal magnetic field

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ABSTRACT

Context. A white dwarf mass appreciably above the Chandrasekhar limit is usually attributed to rotation. A strong internal magnetic field is the other candidate, and the purely toroidal configuration is the one for which the equilibrium problem remains tractable without a Grad-Shafranov solve.

Aims. We quantify how much equilibrium mass a purely toroidal field adds to a cold, non-rotating white dwarf, and how that mass gain depends on the central density, within the density range in which the cold degenerate equation of state can still be claimed to apply.

Methods. The field is prescribed as $B_\varphi = K\rho^m \varpi^{2m-1}$, the form for which the Lorentz force derives from a potential and the barotropic structure equations close. The resulting magneto-hydrostatic equilibria are computed with a self-consistent field iteration in which the density inversion becomes an implicit, per-point root find, and are certified with the virial error and with an explicit test that the configuration fits inside its computational domain.

Results. The dimensionless field strength $E_{\rm mag}/|W|$ is predicted analytically to be independent of central density and radius, and is measured to vary by 0.9% across two and a half decades in ρ_c. A field with $E_{\rm tor}/|W| \simeq 0.18$ raises the equilibrium mass by $\simeq 47\%$, essentially independent of central density, giving $M > 2\,M_\odot$ inside the density range allowed by the inverse-beta-decay threshold, with a maximum interior field of $2\text{–}3 \times 10^{14}$ G. Adding rigid rotation on top of such a field shortens, rather than extends, the range of angular velocity over which the sequence remains virially certified.

Conclusions. A purely toroidal field is sufficient, on its own, to reach $2\,M_\odot$ at physically admissible central densities. It is, however, the configuration most exposed to the Tayler $m = 1$ instability, which the axisymmetric formulation used here cannot test, and it produces no exterior field and hence no dipole for magnetic braking to act on.

Key words. white dwarfs – stars: magnetic field – magnetohydrodynamics (MHD) – methods: numerical

1. Introduction

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2. Equation of state

The background stellar structure throughout this work is that of a fully degenerate, zero-temperature electron gas (ztwd; Chandrasekhar 1935). Pressure and density are both parametrized by the normalized Fermi momentum $x \equiv p_F/(m_e c)$,

$$P(x) = A\left[x(2x^2 - 3)\sqrt{x^2 + 1} + 3\sinh^{-1} x\right], \tag{1}$$

$$\rho(x) = B\,x^3, \tag{2}$$

where A is a fixed combination of fundamental constants (m_e, c, $\hbar$) and B depends on the composition only through the mean molecular weight per electron $\mu_e = 1/Y_e$, with Y_e the number of electrons per baryon. This is a purely mechanical equation of state: at fixed composition, pressure depends on density alone, with no temperature dependence. Consequently, once the configuration is restricted to be non-rotating and field-free, the entire stellar structure (mass, radius, density profile) is determined by exactly two physical parameters: the central density ρ_c and μ_e.

The specific enthalpy has a closed form,

$$H(x) = \frac{8A}{B}\left[\sqrt{1 + x^2} - 1\right], \tag{3}$$

which is what makes the self-consistent field (SCF) iteration of Sect. 4 tractable: in the field-free case the density is recovered directly from the local enthalpy via the inverse relation $x(H) = \sqrt{(1 + HB/8A)^2 - 1}$, with $H \leq 0$ mapped to $\rho = 0$ (the stellar surface). Section 3.2 describes how this step changes once a toroidal field is present.

Near the surface ($x \ll 1$) the equation of state reduces to $\rho \propto H^{3/2}$, an effective polytropic index $n = 3/2$; in the relativistic core ($x \gg 1$) it approaches $\rho \propto H^3$, $n = 3$ – the marginal polytropic index at which the equilibrium mass becomes independent of central density, the origin of the Chandrasekhar mass limit.

At sufficiently high density, inverse beta decay (electron capture) changes the composition locally and the constant-μ_e cold equation of state above ceases to describe a real white dwarf. We adopt the tabulated threshold densities of Boshkayev et al. (2013) for this validity limit. This gate is not a formality here: it is what distinguishes a mass gain that can be claimed for a white dwarf from one obtained at a central density where the assumed microphysics no longer holds, and every configuration quoted as a result in Sect. 5 lies below the threshold for the assumed composition.

3. A purely toroidal field

3.1. The field law

For a barotropic star the magnetic force must derive from a potential if hydrostatic balance is to close in the form of a Bernoulli integral. For an axisymmetric, purely azimuthal field $\mathbf{B} = B_\varphi(\varpi, z)\,\hat{\mathbf{e}}_\varphi$, where ϖ is the cylindrical radius, this restricts B_φ to the one-parameter family

$$B_\varphi = K\rho^m \varpi^{2m-1}, \qquad m \geq 1, \tag{4}$$

with K setting the field amplitude and m its concentration towards the dense inner regions. The corresponding Lorentz force, obtained from the divergence of the Maxwell stress tensor, is $\mathbf{f}_L/\rho = -\nabla M_{\rm tor}$ with

$$M_{\rm tor}(s) = \frac{mK^2}{4\pi(2m-1)}\,s^{2m-1}, \qquad s \equiv \rho\varpi^2, \tag{5}$$

which we verified symbolically for $m = 1, 3/2, 2, 3$ (the residual of the force balance vanishes identically). Hydrostatic equilibrium then reduces to the algebraic Bernoulli integral

$$H + \Phi + M_{\rm tor}(\rho\varpi^2) = C, \tag{6}$$

with Φ the gravitational potential, solved together with the Poisson equation for Φ and with the equation of state of Sect. 2.

Two features of Eq. 4 are worth making explicit, because they set both the appeal and the limits of this branch. First, B_φ is an *algebraic* function of the local density: no flux function, no Grad-Shafranov operator, and no Green's function are involved in obtaining the field, in contrast with the poloidal case. Second, and for the same reason, the field vanishes wherever the density does. A purely toroidal field is strictly confined to the stellar interior; there is no exterior field and no dipole moment, a point we return to in Sect. 6.

3.2. The implicit density inversion

Because $M_{\rm tor}$ depends on ρ – the very quantity being solved for – the density update of the SCF iteration is no longer a direct inversion of the equation of state. At each grid point one must instead solve

$$H(\rho) + M_{\rm tor}(\rho\varpi^2) = C - \Phi \tag{7}$$

for ρ. The left-hand side is monotonically increasing in ρ (both H and $M_{\rm tor}$ are), so the root is unique and safely bracketed, and we obtain it with a standard bracketing root finder. This is a change in the structure of the algorithm, not in its physics, and it is the only place where the toroidal branch is more expensive than the field-free one.

3.3. Virial diagnostic

Every configuration is certified with the scalar virial identity, which for a static magnetized star reads

$$W + 3\Pi + \mathcal{M} = 0, \qquad \mathrm{VE} \equiv \frac{|W + 3\Pi + \mathcal{M}|}{|W|}, \tag{8}$$

with W the gravitational energy, $\Pi = \int P\,dV$, and $\mathcal{M} = \int B_\varphi^2/8\pi\,dV$ the magnetic energy. For the field law of Eq. 4 the magnetic contribution satisfies

$$\int \rho\,\nabla M_{\rm tor}\cdot\mathbf{r}\,dV = -\int \frac{B_\varphi^2}{8\pi}\,dV, \tag{9}$$

the sign following from $M_{\rm tor}$ entering Eq. 6 with a positive sign, combined with the convention-free Maxwell-stress identity $\int \mathbf{r}\cdot\mathbf{f}_L\,dV = +\int B^2/8\pi\,dV$, valid for any purely toroidal field. We confirmed this numerically: the residual of Eq. 8 falls monotonically, 2.27% $\rightarrow$ 1.00% $\rightarrow$ 0.36% $\rightarrow$ 0.14%, as the radial mesh is refined from $n_r = 65$ to $n_r = 257$. We take VE $< 10^{-3}$ as the acceptance threshold throughout, and quote it for every configuration reported below.

4. Numerical method

Equilibria are obtained with the self-consistent field scheme of Hachisu (1986), in which the density field, the gravitational potential and the Bernoulli constant are updated in turn until the density field stops changing. The gravitational potential is obtained from a Legendre (multipole) expansion of the Poisson Green's function, truncated at $l_{\rm max}$; the magnetic term enters only through Eq. 5 and the implicit inversion of Sect. 3.2. Three numerical issues proved to matter quantitatively for this branch, and we record them because each one, on its own, is capable of producing a wrong physical conclusion.

Domain size. A toroidal field makes the star *prolate* ($R_{\rm pol} > R_{\rm eq}$), the opposite of the oblate deformation produced by rotation, and the deformation grows with K. A computational domain sized from a field-free radius estimate is therefore progressively too small as K increases, and the "surface" the solver finds is the edge of the domain rather than a physical radius. We monitor this directly through $f_{\rm pol} \equiv R_{\rm pol}/r_{\rm max}$ and accept only configurations with $f_{\rm pol} \lesssim 0.2$; a value of 1.0 means the star does not fit in its box at all. This is not a conservative margin added for safety – an earlier pass of this work concluded that $2\,M_\odot$ was *not* reachable, a conclusion produced entirely by domain overflow at every point of the sequence with the strongest field, and reversed once the domain was enlarged. Domain size and mesh resolution must be varied one at a time: enlarging the domain at fixed Δr drops VE from 1.43×10^{-2} to 2.49×10^{-4} and then leaves M and VE unchanged to four significant figures under a further enlargement, whereas refining the mesh inside a domain that is too small does not improve VE at all.

Continuation in K. Direct solves at $K \geq 5\times10^{-3}$ do not converge at any domain size we tried. These configurations are reachable only by continuation: stepping K upward in small increments from the field-free solution and warm-starting each step from the previous one.

The multipole expansion. Forming the radial Green's function as separate powers $r'^{\,l+2}$ and $r^{-(l+1)}$ before dividing overflows double precision outright at $l = 48$ for stellar radii, and silently loses precision even at $l_{\rm max} = 16$, where the two factors are separated by some 270 orders of magnitude before a division intended to cancel most of that scale. Rewriting the radial integrals as a recursion that only ever forms the ratios $(r_{i-1}/r_i)^{l+1} \leq 1$ and $(r_i/r_{i+1})^l \leq 1$ – the same integral, with no absolute power of r

Table 1. Equilibrium mass with and without a purely toroidal field (m = 1, $K = 3 \times 10^{-3}$, no rotation), at three central densities. All points are certified at $f_{\rm pol} \simeq 0.09$–0.16 and reached by continuation from $K = 0$.

ρ_c ($\mathrm{g\,cm^{-3}}$)	$M(K=0)$ ($M_\odot$)	$M(K)$ ($M_\odot$)	gain	$E_{\rm tor}/\lvert W\rvert$	VE
1.0×10^{10}	1.4108	2.0722	+46.9%	0.1809	4.0×10^{-4}
1.5×10^{10}	1.4159	2.0874	+47.4%	0.1805	3.9×10^{-4}
1.0×10^{12}	1.4323	2.1426	+49.6%	0.1790	2.7×10^{-4}

Notes. The $\rho_c = 10^{12}\,\mathrm{g\,cm^{-3}}$ row is given for reference only: it lies above the inverse-beta-decay threshold (Sect. 2) and is not a physically admissible white dwarf.

ever formed – removes the problem. With this in place, the truncation itself is not a limitation for the configurations considered here: stepping $l_{\max} = 16 \rightarrow 32 \rightarrow 48$ moves VE by less than 10%.

5. Results

5.1. An invariance predicted before it was measured

For $m = 1$ the field law is $B_\varphi = K\rho\varpi$, and the dimensionless field strength can be estimated without solving anything. With $E_{\rm mag} \sim K^2\rho^2R^5/8\pi$ and $|W| \sim GM^2/R \sim G\rho^2R^5$, the density and radius cancel:

$$\frac{E_{\rm mag}}{|W|} \sim \frac{K^2}{8\pi G}, \tag{10}$$

independent of both ρ_c and R, up to a shape factor of order unity that drifts slowly along the sequence as the effective polytropic index slides from 3/2 towards 3. Equation 10 predicts that the fractional mass gain produced by a given K should be essentially the same at any central density – and, for $K = 3 \times 10^{-3}$, that it should be of order 50%.

Table 1 shows the measured values at three central densities spanning two and a half decades. The prediction was made before these runs and is confirmed by them: $E_{\rm tor}/|W|$ varies by 0.9% over the full range, and the fractional mass gain by less than three percentage points.

5.2. The mass gain

The result is best stated in terms of $E_{\rm tor}/|W|$ rather than K, since K carries the units and the arbitrariness of the ansatz of Eq. 4 and means nothing outside it, whereas the energy ratio transfers to any other field configuration: a toroidal field with $E_{\rm tor}/|W| \simeq 0.18$ raises the equilibrium mass by $\simeq 47\%$, essentially independent of central density, which puts $M > 2\,M_\odot$ inside the density range the inverse-beta-decay threshold allows. The two admissible rows of Table 1 both exceed $2\,M_\odot$ while remaining below the neutronization threshold for $\mu_e = 2$; the mass gain does not have to be bought by pushing ρ_c into a regime the equation of state cannot describe.

In physical units, the corresponding interior fields at those two points are

$$B_{\varphi,\max} = 1.95 \times 10^{14}\ \mathrm{G} \quad \text{and} \quad 2.56 \times 10^{14}\ \mathrm{G},$$

with volume-weighted means of 7.9×10^{12} and 9.9×10^{12} G respectively. These are the field strengths that come out of the calculation, with no tuning towards any target value.

5.3. Rotation does not extend the sequence

Since a toroidal field elongates the star and rotation flattens it, a natural expectation is that combining the two should push the mass-shedding limit to higher angular velocity than rotation alone. We tested this directly, running the same continuation in the central angular velocity Ω_c twice on an identical mesh and at identical ρ_c, once field-free and once with a moderate toroidal field ($K = 3 \times 10^{-3}$, $E_{\rm tor}/|W| \simeq 0.18$ at $\Omega_c = 0$).

The expectation fails. With the field present, the equatorial mass-loss ratio and the virial error degrade *together* and smoothly between $\Omega_c = 15$ and $20\,\mathrm{rad\,s^{-1}}$: the mass-loss ratio crosses unity, and VE crosses the 10^{-3} acceptance threshold at essentially the same Ω_c and then climbs monotonically to 0.16 by $\Omega_c = 32\,\mathrm{rad\,s^{-1}}$. The practical limit is therefore $\Omega_c \simeq 17$–$18\,\mathrm{rad\,s^{-1}}$, about a third lower than the $\simeq 26\,\mathrm{rad\,s^{-1}}$ reached in the field-free case. The simultaneous, progressive failure of two independent diagnostics is the signature of a genuine loss of virial balance rather than of the solver jumping to a spurious branch. The +47% of Table 1 should therefore be read as a static ceiling for the toroidal branch on its own, not as a floor to which rotational support can simply be added.

6. Limitations

Three limitations bound what the results above can be taken to mean, and none of them is incidental to the method.

No stability test. These are two-dimensional axisymmetric equilibria, certified to satisfy hydrostatic and virial balance. A purely toroidal field is precisely the configuration subject to the Tayler $m = 1$ instability (Tayler 1973; Markey & Tayler 1973), which is non-axisymmetric and therefore invisible to this formulation by construction. Nothing here tests whether the configurations of Table 1 survive a non-axisymmetric perturbation, and the expectation from stability theory is that, in this pure form, they do not. A companion investigation addresses this by relaxing a random seed field dynamically in three-dimensional MHD and measuring which mixed poloidal-toroidal configuration survives, rather than imposing one.

No exterior field. By Eq. 4 the field vanishes with the density, so it is confined to the stellar interior and the exterior solution is field-free. There is no surface dipole, and hence nothing for magnetic braking to act on – which makes this branch, on its own, unable to represent the rotational evolution of a magnetized white dwarf, however well it represents its equilibrium structure.

An unresolved corner of parameter space. At $K = 5 \times 10^{-3}$ mesh refinement does not converge cleanly: VE falls from 8.2×10^{-3} to 9.6×10^{-4} and then rises again to 3.7×10^{-3} at the finest mesh tested, a non-monotonicity that survives the Green's-function fix of Sect. 4 and that we have not been able to attribute either to a genuine sensitivity near a marginal equilibrium or to a further numerical effect. We flag it as open rather than resolve it in either direction. It does not affect $K = 3 \times 10^{-3}$, the value carrying the results of Sect. 5, which converges monotonically under mesh refinement ($1.5 \times 10^{-2} \rightarrow 1.7 \times 10^{-3} \rightarrow 2.7 \times 10^{-4}$) and is unchanged to four significant figures under domain enlargement.

7. Conclusions

TBD.

Acknowledgements. TBD.

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